LITTLE BLUE BOOK NO. 1471
Edited by E. Haldeman-Julius

How to Become Mentally Superior

What Makes One Rise Above the Average

Henry M. Le Chatelier

(Translated from the French by Ralph E. Oesper)

LITTLE BLUE BOOK NO. 1471
Edited by E. Haldeman-Julius

How to Become Mentally Superior

What Makes One Rise Above the Average

Henry M. Le Chatelier

(Translated from the French by Ralph E. Oesper)

Title: How to become mentally superior
What makes one rise above the average

Author: Henri Le Chatelier

Editor: E. Haldeman-Julius

Translator: Ralph E. Oesper

Release date: July 1, 2026 [eBook #78988]

Language: English

Original publication: Girard: Haldeman-Julius Publications, 1929

Other information and formats: www.gutenberg.org/ebooks/78988

Credits: Tim Miller, Sam Lamb and the Online Distributed Proofreading Team at https://www.pgdp.net

LITTLE BLUE BOOK NO. 1471

Edited by E. Haldeman-Julius

How to Become Mentally Superior

What Makes One Rise Above the Average

Henry M. Le Chatelier

(Translated from the French by Ralph E. Oesper)

HALDEMAN-JULIUS PUBLICATIONS

GIRARD, KANSAS

PRINTED IN THE UNITED STATES OF AMERICA

HOW TO BECOME MENTALLY SUPERIOR

Every school gains renown not only through the scientific achievements of its professors, but also because of the industrial successes of its former students. Schools have been a potent factor in the development of an intellectual elite, the class responsible for the progress of civilization in any country. If Europe is superior to Africa, the sole cause lies in the possession of leaders. The blacks of savage countries may be good manual laborers, but they lack a select class to direct them, either as governing officials, as officers in warfare, as scholars, as engineers or as organizers of their industries. The formation of an intellectual superior class should be the dominant preoccupation of any country that expects to cut a figure in world affairs.

The geologist, de Lapparent, in a didactic statement declared that every terrain is, of necessity, divided into three strata; the upper, the middle and the lower. The intellectuals likewise may be placed on three levels; the men of genius, whose fame and influence extend throughout the world for many centuries; the great men, whose renown, however great at a given time, is finally eclipsed by that of their successors, and lastly, the lower elite, who temporarily exert a useful influence within rather narrow boundaries, but never attain far-reaching notability. Each of these three categories of intellectual superiors renders about the same value of service to humanity; the men of genius are certainly the greatest benefactors, but they also occur most seldom. In algebraic terms, the product of the number in each class multiplied by each individual's usefulness gives a constant.

In a talk to the students of an American university Carnegie said: "I am speaking only to those of you who are ambitious to become millionaires; the others do not interest me." The present speaker wishes to emphasize a parallel thought: "I am speaking only to those of you who have an ambition to raise yourselves above the average, and I believe this will include all of you." It would surely be folly for anyone to deliberately set out to become a genius, because this goal can only be reached through exceptional qualities, but we all can and should strive to be numbered among the elite, to use this term in its proper sense. With the exception of certain afflicted individuals, fortunately not numerous, all of us from birth have the requisite qualities.

The rest is dependent on will power and on the method of developing and applying our natural endowments.

Let us examine together, using the experimental method, the conditions attendant upon the recruitment of the intellectual elite. For this purpose we need not distinguish the levels of attainment, for they do not differ in nature, but only in degree. We can then cite as examples great men with whose lives you are more familiar, and from these we may draw conclusions applicable to the formation of the ordinary elite. What qualities are essential and how may these be developed?

ACTIVITY

The most striking characteristic of great men is their zeal for work. None of them observed the eight-hour day, no matter what the field of their activities. We may cite as examples great statesmen, such as Napoleon or Louis XIV; great writers, such as Victor Hugo or Lamartine; great artists, such as Michelangelo or Leonardo da Vinci; great scientists, such as Lavoisier or Pasteur; great manufacturers, such as Bessemer or Siemens. In truth, they often employed the most varied ruses to protect their working periods from interruption. Napoleon assembled his ministerial council during the soirees at the Tuileries, leaving the reception of guests to the Empress Josephine. Buffon took refuge in his country house and there peacefully wrote his natural history. Descartes secreted himself in a little Dutch village when he desired to cultivate his philosophical meditations. The labor expended by celebrated men is sometimes greatly underestimated. Powers of extemporaneous speaking far beyond reality are often ascribed to great orators. As a matter of fact, the most successful of them write out their addresses in full before delivering them. Mistaken notions as to this have originated from false claims. Emile Zola pretended that his voluminous literary output required only three hours’ daily toil. Perhaps he did not actually keep the pen in his hand longer than that, but the final wording comprises only a small part of literary production.

Francisque Sarcey, while discussing the art of lecturing, very judicially analyzed the importance of preliminary work. He said:

“The title of a lecture should be chosen a month in advance of the delivery; then the subject matter should be considered for two weeks during every free moment, especially while strolling about. By degrees, new and interesting points of view will appear spontaneously; these should be classified either in the memory or jotted down systematically. During the third week, the material thus accumulated should be gone over mentally, the less important points rejected or suppressed, the others rearranged in their logical order and the connecting thoughts brought to light. At last, during the fourth week, the final wording is committed to paper and this requires no great effort.”

Great men have not only labored much, but their efforts have been confined to a few specialties, thus increasing the intensity of their work. In hydrostatics a force is concentrated on a piston of small area in order to produce great pressures. Saint Claire Deville devoted half of his career to the study of dissociation. Berthelot worked fifteen years on organic synthesis, fifteen years on thermo-chemistry and fifteen years on agricultural chemistry. Many scientists owe their fame to studies made in a single field as instanced by Pasteur in microbiology, Fresnel with the theory of light, Ampere and the laws of electrodynamics. The same holds true in industrial applications and as examples we have Vicat and hydraulic cements or Fourneyron and the turbine.

This concentration of effort cannot be recommended too highly to young investigators, for they frequently exhibit an opposite tendency and allow themselves to be enticed from one thing to another by topics which appeal to them. Only men of exceptional endowments, like Leonardo da Vinci or Lavoisier, can successfully distribute their efforts without paralyzing their creative powers. Some scholars carry this specialization of their endeavors to excess and pride themselves on the extent to which they disregard the obligations of daily life. Many stories in this vein are related of Ampere and of Henri Poincare. The following actual occurrence illustrates the same point. I was invited to dine with an illustrious foreigner and on arriving at the hotel I was told by my host that his wife was ill and consequently she could not dine with us. He said: “Under these conditions will you be kind enough to order the dinner, for since I have never studied this subject, I know nothing about such matters.”

It is not sufficient to work hard, but is also essential to work efficiently, i. e., time must not be wasted on useless projects. A plan of attack should be formulated in advance of starting the actual work of writing, so that there need be no hesitation. Attempts to do two things at the same time are usually fruitless, and it should be matter of principle not to stop working until something definite has been achieved. Learn to persevere and do not hesitate to adhere to a decision made after proper reflection. It is this spirit of organization, this convergence of efforts that is so highly manifested by great political leaders such as Louvois, Napoleon, Cavour, Mussolini.

Much gain may accrue by organizing the vague, spontaneous thoughts which the mind cannot suppress, even though they appear to have little value. We are always thinking about something, and this involuntary thought is much less fatiguing than mental effort consciously directed toward definite production. This preparatory reflection is sometimes erroneously regarded as being quite distinct from the real work, but this opinion is quite wrong, for preliminary thought is an essential forerunner of all creative achievement. In fact, it is just as indispensable as the final effort and the latter will certainly be of little avail if the way has not been properly prepared. If the mind could be trained not to think useless thoughts, the productive capacity would be enormously enhanced. When Newton was asked how he had discovered the laws of universal attraction, he replied: “By always thinking about them.” This may be the dominant reason for the superiority of great men, but we really know very little about this fugitive thinking, whose manifestations are not external. In fact, the originators of such mental processes are sometimes not conscious of their operation, or, as we say, we are here dealing with the subconscious. Henri Poincare claimed that he thought during sleep, and on waking would find at hand the solution of problems which had baffled him the day before. However, this is not a commendable practice, for it is opposed to the rest which each night’s sleep should bring.

How may a zeal for work be developed? Is it a natural gift or is it a result of education? The greatest stimulant of activity is habit, which proverbially becomes second nature. After leading an active life, it is not possible to stop work without suffering. Idleness due to retirement rapidly kills many men who previously had enjoyed excellent health. After the habit of working is once formed, a man will work for the mere joy of working just as we walk for the pleasure of the exercise. It has become a necessity.

However, this habit is not easily acquired. Temperament plays some part. Certain children, from birth on, exhibit more will power, have more acute faculties of attention, are more persevering, all of which are essential to the accomplishment of a protracted task. Yet these predispositions are, in general, developed only to a slight degree and play only a minor part in the differentiation of individuals. Other factors seem to be of greater importance.

The example of the home and of companions exercises a preponderant influence. A child who all his life has seen an industrious father will merely through imitation be led to accept the law of the obligation to work. Pascal, Lavoisier, Pasteur were raised in families in which honor was paid to industry. Whether the latter is intellectual or manual matters little. Very few, or perhaps no great men, have come from the families of the idle rich.

A second very potent factor is ambition, that is the desire to acquire riches or honors. Men not favored by the fortunes of birth sometimes struggle with extreme energy to make a place for themselves. A striking instance of the power of ambition is found in the career of Senator Leopold Goirand, who died recently. He published some essays on education which reveal curious points in his psychological makeup. At the age of fifteen, he conceived the dual ambition to become very rich and to attain a powerful political position. He succeeded in both endeavors. For twenty years he forced himself to be content with six hours of sleep each night in order to lengthen his working day. Each morning on arising he spent two hours acquiring general culture, the rest of the day was devoted to his business, and finally the evenings were passed in attendance on social affairs, for the latter are extremely useful in the prosecution of a career. Not until his physician warned him that he was no longer fit to continue this program did he consent to sleep eight hours nightly. Many similar examples may be cited.

In Bessemer's autobiography, which is a veritable romance, he tells of his superhuman efforts, as a young man, to earn enough money to marry. While Cavour was striving to create the Italian kingdom, he allowed himself only five hours' sleep each night so that he might have time for the stupendous task whose realization had been the dream of his whole life. He took over the direction of four ministries at one time.

A third stimulus, more noble than those already discussed, is the attraction inherent in the fruits of labor, i. e., the joy of knowledge and the pleasure of performance. The passion for knowledge or success in a chosen field often arouses men who by temperament or habit might have been inclined to loaf. A pertinent example is Mallard, one of the scientific glories of France. Like many others who graduated from the Ecole Polytechnique at the top of the class he seemed to be destined for a standardized, peaceful career in the governmental service. As engineer at Gueret and then as professor at Saint Etienne he divided his activity between long journeys and everyday affairs, attending to his ministrative duties and his teaching. At the age of forty he was appointed professor of mineralogy in the School of Mines in Paris and consequently because of his teaching duties he found himself obliged to study this science. He became deeply interested in one of its branches, crystallography, and for twenty years, until his death, all his efforts were concentrated in this field. He succeeded in working out original demonstrations of the laws of crystallography and he created a new chapter in this field, the theory of crystalline groupings.

Many similar cases are found among scholars, for many of them are motivated chiefly by the joy of knowledge. On the other hand, examples of this disinterested activity are less frequent among industrialists. However, the pleasure of achievement, rather than the mere love of gain, has actuated the greatest of these. The optician Zeiss worked for the glory of his country, Germany, and his native city, Jena. The Danish brewer Jacobsen engaged in business only to gain the means of endowing the museums and laboratories of Copenhagen, which have become world famous. The American millionaire Carnegie while a young man spent his free time in libraries solely because of his desire to learn. He later devoted the major portion of his immense wealth to the development of public libraries and to the founding of an institute of scientific research. Henry Ford left the farm and worked in a locksmith's establishment because of a desire to learn the use of tools, and even now he derives great pleasure from heading a well-organized industry. I knew two contractors who had taken part in the construction of the Suez Canal. They retired from business and took up agriculture. One engaged in stock-raising, the other developed a model farm. They devoted all their energies to the enterprises and ran them so that the receipts and expenses balanced, neither profit nor loss resulting. Their sole ambition was to do a good piece of work and to turn out products of superior quality.

This joy in work may be developed by education and without difficulty. Success is assured if less attention is given to preparing for examinations and more stress laid on the intellectual molding of the children. From their earliest years they have a wide-awake curiosity, they continually ask why and how. Instead of eradicating this disposition, it should be cultivated. Science courses lend themselves wonderfully to this end. Emphasis should be placed on the linking together of facts, which is the essence of the scientific method, discarding the fastidious enumeration of isolated facts which so overburden the memories of pupils today. Any child can be interested in the consequences of Pascal's laws of hydrostatics and made anxious to work diligently to understand them. A study of Archimedes' principle, applied to floating bodies, of hydrometers, of water levels in connecting vessels, of atmospheric pressure—all these may be grouped around Pascal's laws and thus made into an attractive ensemble. It is wrong to treat each of these topics as a separate chapter as many physics texts do, for then the intimate relationships disappear from view and likewise all attractiveness is lost.

Manual exercises should be included in the secondary curriculum because the formation of ideas in young people is rendered more easy and pleasant by combining sight and touch. They love motion. For instance, when teaching hydrostatics, the pupils may be directed to cut cubes from various kinds of wood, to measure them, to weigh them and finally to determine the loss in weight when the cubes are immersed in water. An exercise of this kind makes the appreciation of Archimedes' principle pleasurable. In the same way, preliminary exercises in graphic plan-making lead to a much easier understanding of geometrical reasoning. Much less intellectual effort is required to comprehend the demonstration of a truth if actual experimentation has previously made the reality familiar.

A final incentive to the ardor for work is good health. The thought of working or still more of getting to work, i. e., the wish to do something, involves, if not a true fatigue, at least a feeling of fatigue which leads many to shrink back. A good digestion and restful sleep make the thought of work much more agreeable. This does not imply that a strong will cannot overcome the weakness arising from poor health, for there are remarkable instances of such victories, but they are rather exceptional. Pascal subdued his infirmities, but the strain finally killed him. Health always is a great driving power, and the truth of the axiom, "*mens sana in corpore sano*," cannot be debated. Physical culture should occupy an important place in the education of the young; it is indispensable in the formation of the intellectual elite of a nation. However, it must not be forgotten that muscular fatigue renders all mental labor impossible for the time being. The physical exercise should follow intellectual exertion, but should never precede. While in Holland, Descartes devoted his mornings to philosophy and cultivated his garden in the afternoons.

IMAGINATION

A useful member of society does not merely work hard and produce much for his own benefit; he should add to the common fund of knowledge. In other words, he must produce new ideas, discover scientific laws, devise new literary or artistic presentations, perfect methods of government; in short, he must play a creative role.

What is the mechanism by which this progress is realized? Contrary to popular belief, our knowledge does not increase by leaps and bounds, but the development is regular and very slow. Each step forward, in the majority of cases, comes from the simple combination of facts previously known. It is only necessary to delve in the storehouse of knowledge and to bring new relations to light. This correlation is a fruit of the mental faculty, imagination, whose functioning is rather capricious. The solution of a problem may be sought unsuccessfully for a long time, and then suddenly it may flash into the mind at a time when the problem is no longer being consciously considered.

The work of all great men shows the employment of imagination; it constitutes the beginning of all great discoveries. Pascal created hydrostatics by connecting the limitation of the height of a barometric column with the weight of the atmosphere; Newton discovered universal gravitation by comparing the movement of the planets with the fall of an apple; Pasteur founded microbiology by connecting the spread of disease with the life processes of microscopic organisms. Likewise, in the field of letters, we find writers dealing with ideas common to all humanity and sending them forth in new attire. La Fontaine gave life to Aesop's fables by endowing the animals with speech; Corneille introduced a sense of duty magnified almost to heroism into the Spanish dramas. Artists sometimes slightly exaggerate certain features of their models to produce a more striking representation. Michelangelo accentuated the muscular development of heroic figures and Raphael the grace of women.

The same is true in industry. Sir William Siemens applied the theoretical reasoning of Sadi-Carnot to the heating of hearths, and the regenerative furnace resulted. He invented neither thermo-dynamics nor industrial heating, but by uniting these two sets of facts, he brought about a great advance, which made possible the modern processes of making steel in open hearths and of glass in tank furnaces. All inventors possess this mental activity, sometimes to excess. It is interesting to read Bessemer's autobiography from this point of view. We see his mind always in feverish agitation, trying each day to produce something new, usually without success.

This first type of imagination is meditative. It acts slowly, and to a certain degree may be governed by the will. There is a second type, rather more delicate in nature, whose action is sudden and not preceded by reflection. It is this type which enables us to see at a glance all the correlations and distant consequences of a chance observation. The predisposing factors are impressionability and nervous sensibility. This quality varies greatly from individual to individual, certain minds respond to the slightest external suggestion, while others feel nothing, see nothing. In general, great scholars are characterized by a highly developed aptitude for sensing and using facts presented to them. The accidental observation that his determinations of the density of nitrogen were discordant led Lord Rayleigh to the discovery of argon. Other investigators had been faced with the same phenomenon, but were not markedly impressed. While attempting to fuse platinum, Saint Claire Deville was struck by the difference between the calculated temperature of the oxy-hydrogen flame and the observed melting point of platinum. This led him to suspect the dissociation of water vapor, while his collaborators, possessed of the same facts, thought nothing of them. Similarly, Bessemer was led to his process of producing steel in a converter by the fortuitous observation of the formation of malleable steel during an attempt to harden cast iron. Or better still, Auer von Welsbach discovered the incandescent mantle through a chance observation of the light emitted on calcination of precipitated thoria. Numerous analytical chemists had doubtless seen the same thing, but their attention was not arrested.

Great artists viewing nature, great generals before a battle, great lawyers before a trial are sensible to instantaneous impressions which escape the notice of ordinary men. The two types of intellectual activity seem to be predominantly natural endowments. Some children have wide-awake minds, others are more dull and remain thus throughout life.

This quality may, however, be developed by education, and more attention should be paid to this phase of education, than is usually the case. Exercises in writing composition, problems in geometry, afford excellent material from which to build up mental habits of using accumulated knowledge or of seeking new correlations of known facts. This type of training is doubtless the most useful function of secondary education. On the other hand, education cannot develop the second type of mental activity which is not dependent on reflection, but which functions instantly in some way not known to us. Nevertheless, there seems to be some possibility of perfecting this natural endowment by suitable laboratory exercises.

JUDGMENT

The combining of imagination and work, i. e., intellectual activity joined to bodily activity, does not entirely suffice for the making of a great man. Excellent examples are inventors, who almost without exception are possessed of active minds and an equal ardor for work, and yet few of them become great. In fact, many of them have difficulty in making both ends meet, and they remain mediocrities. The two qualities can only be used efficiently if joined with a third; namely, common sense. This latter, if developed to its highest degree, becomes what Pascal has called the sense of finesse. Bodily and mental activity are certainly powerful instruments, but like all aids they must be used judiciously. Common sense should guide the choice of problems to be studied.

One of the most potent reasons for the success of great men is that they were wise enough to apply their efforts to worth-while problems. Why will the names of Lavoisier, Sadi-Carnot, Ampere, Fresnel, Saint Claire Deville, Berthelot always be famous? It is because of the greatness of the topics they studied. The results of their discoveries in chemistry, thermo-dynamics, electrodynamics, physical optics, chemical mechanics, organic synthesis have echoed again and again, and the reverberations are daily multiplied.

Many years ago Taine said that the systematic study of the dominant features of his subject was the essential characteristic of the work of a true artist. The same holds true in all realms of human activity. There are dominant phenomena whose influence is felt under most manifold circumstances. A knowledge of these favoring factors is of inestimable value to the human race, and their discoveries merit suitable recognition on the part of their fellowmen.

A second form of judgment is “critical sense,” indispensable to both scholars and to those directing industries. This gift makes possible the detection of errors in measurements or leads to a premonition against erroneous interpretations of observations. It is often lacking in inventors, who are prone to persist in their notions despite self-evident failures. Lesser intellectual lights also cannot bring themselves to abandon favorite hypotheses which are not in agreement with the facts; they seek refuge in new additional hypotheses and cling fast to the original notion. Ditte, a pupil of Saint Claire Deville, furnished a striking example of this type of mental gymnastics. While trying to extend the law of fixed dissociation tensions to the decomposition of solutions of mercuric sulphate, he found that the concentration of the liquid with respect to sulphuric acid increased with increasing concentration of mercuric sulphate. Instead of abandoning his hypothesis, which was obviously inaccurate, he introduced a second, which today appears absurd, but for a time it enjoyed a certain credence among chemists. He postulated that the dissolved mercuric sulphate was present in two forms, part as neutral salt and part as basic salt, dissolved in, but not combined with, sulphuric acid. He calculated the proportion of the basic salt supposedly thus dissolved which would leave a constant concentration of sulphuric acid in solution, and he viewed the results of this arbitrary calculation as experimental verification of his hypothesis. This species of error is constantly perpetrated nowadays by those who speculate as to the constitution of matter. Savants who, like Lavoisier, Claud Bernard, Pasteur, combine an ever-watchful imagination with a critical sense severe enough to lead them to discard hypotheses found contrary to fact are extremely rare.

Finally, there is a still more refined form of common sense, namely, the sense of subtle discrimination which enables us to guide our minds directly into domains of thought which are not perfectly obvious. It is often asserted that hypotheses are free to all, an investigator may postulate what he pleases, provided he finally subjects his notions to precise, experimental test. However, it is distinctly worth-while not to set up too many inexact hypotheses, for time should be economized to the end that production may be increased. A certain instinctive discernment is necessary to set one rapidly on the best line of procedure. Rules for this cannot be laid down; it is a matter of feeling and not of reason. The remarkable productivity of Pasteur was doubtless due to his ability, from the very first, to thoroughly organize his researches. Sometimes this ability is ascribed to chance, but this is not correct, for we see here the fruit of a very shrewd form of common sense. Saint Claire Deville’s thought that there might be a possible analogy between the phenomena of decomposition and those of vaporization was an intuition of genius; it led him to the discovery of chemical equilibrium, which he termed dissociation, and a new science, chemical mechanics, has been erected on this idea.

Good common sense is often a gift of nature, but the more delicate sense of subtle discrimination is principally a result of education. It is very rarely observed among the children of the lower grades; it is a product of classical education, and, above all, it springs from that which is taught in the home. The English declare that thirty-six years of education are necessary to make a gentleman, twelve for the grandfather, twelve for the father and twelve for the son. The same holds true for this sense of finesse. Pascal, Lavoisier, scholars of the first rank, came from families of long-standing culture, and their successes were due in large measure to the prolonged efforts of their ancestors.

The study of classics and humanities aids in developing this trait. Literary or historical criticism requires a constant evaluation of opposing points of view to determine the part played by each in domains not amenable to exact measurement. On the other hand, the study of science develops the geometrical viewpoint, i. e., the use of syllogism, which is utterly useless when comparing phenomena possessing no common measure or such as are based on mere probabilities. The exclusive use of rigorous reasoning and an absolute faith in his conclusions are sometimes very dangerous to a savant. They hinder him from taking account of the real value of the hypotheses which he has made and from recognizing the possible errors in his experiments. These modes of thought are not less hazardous to the industrialist to whom they may impart an unwarranted confidence in the predictions as to the advantages of a new business venture or of a new method of manufacture.

DOCUMENTATION

Many of the essentials for laying claim to a right to be numbered among the intellectual elite have been discussed above, but not all. Suppose that a savage has had from birth all the qualities which we have just reviewed, but that he is entirely ignorant of the progress of the science and industry of the civilized world. It would be extremely difficult for him to advance our knowledge, for he knows nothing about such matters. He can accomplish feats which to him seem extraordinarily difficult, such as, cutting flint or extracting iron from ores, just as his ancestors did. To us, however, he would seem ignorant and no one would dream of classing him as a great man.

No one can make innovations or improve our knowledge unless he is cognizant of actual conditions. There are several reasons why this is so. In the first place, one can obviously improve only those things which he really knows. A frequent cause of the failure of inventors is that they knowingly venture into unfamiliar fields. Bessemer, the son of a metallurgist, advanced the science of metallurgy, but made a miserable failure when he attempted to build a large telescope and also when he tried to construct a boat designed to prevent seasickness, because he knew little or nothing about the theory of optical instruments or of mechanical principles.

While serving on a commission appointed to investigate fire damp, I met a physician, intelligent, possessed of a good practice, but obsessed by the demon of invention. Much affected by an explosion in which several hundred miners had been killed, he thought it would be a good thing to send a current of air through the mine to sweep out the explosive gases and thus obviate such disasters. He disclosed his plan to an official in the Department of Public Works, and the latter warmly congratulated him on his initiative. This unfortunate encouragement led him to abandon his practice for a year and he devoted himself to the construction of a ventilating system. We were obliged to inform him that every coal mine in the world is ventilated. Furthermore, twenty kilometers from his home he could have seen in action immense installations which closely resembled the blowing apparatus which he had worked out, and this device differed but slightly from the bellows used for several thousands of years by savages for melting metals.

A second reason for being well acquainted with the field arises from the fact that all creative advances, all discoveries are, for the most part, the result of combining facts already known. The progress achieved by a single individual is, in general, extremely little, but among these short steps forward, one perhaps, like the last drop which causes the vessel to overflow, may make an invention realizable or it may alter the orientation of our scientific ideas.

Pasteur did not invent the communication of diseases, for this was known to all physicians, nor did he discover the existence of living microscopic organisms, which had been studied from the time of Spallanzani. He merely compared these two sets of phenomena and recognized their relations one to the other. Had he not known the facts, he could not have made the discovery. Likewise, Lavoisier did not invent the balance; it had often been used before. Every alchemist who extracted metals from ores had checked the efficiency of his procedures by weighing. On the other hand, all the physicists knew that gases had weight. Lavoisier simply had to combine these facts to recognize that the increase in weight of metals when calcined in the air was due to the absorption of a gas, oxygen. The time was ripe for this discovery, and Lavoisier had only to pluck the ripened fruit.

The same holds true in industry. The open hearth process of making steel resulted from a combination of Reaumur's century-old work on the refining of cast iron and Siemens' new method of heating. In his invention of the incandescent gas mantle, von Welsbach started with Clamond's magnesia "basket" and perfected this device by substituting thoria for the magnesia. This linkage of known facts is so common that no industrial invention is so novel that the lawyers cannot find authorities for prior claims. For the same reason, the defamers of great scholars, great writers and great painters find great pleasure in accusing them of plagiarism. Fontaine has been reproached with having copied from Aesop, and Corneille is accused of servile imitation of Guillen de Castro.

If the popular saying, "There is nothing new under the sun," is surely not strictly true, nevertheless it is quite accurate to state that progress due to any one man is comparatively small. Humanity moves forward at a slow pace, but the thousand-fold accumulation of short steps forward has altered the face of the world. This progressive revolution has only been possible because men have had a thorough knowledge of the accomplishments of their predecessors.

A third reason for being well versed is that accomplishment of anything new demands a knowledge of the technique of the field, and this has to be learned. If Bessemer had not been a founder in his youth he would never have invented his process of making steel. A thorough knowledge of analytical chemistry is essential to the making of chemical discoveries; success in literature is only possible to one who really knows his own language; a painter must know how to draw. This assertion may appear to be a step backward. Too many artists try to paint, knowing neither drawing nor manipulation of colors; too many scientists have no interest in methods involving actual measurements, but they are content to construct their science with pencil and paper only. These varieties of modern painting and science are very fruitful for their devotees. However, who dares to assert that the cubists and similar faddists will some day be classed as great painters or that much of our present-day theories of the constitution of matter will be highly regarded fifty years from now. In comparison, the men who discovered new laws in chemistry, electricity, optics, etc., will be just as honored thousands of years hence, as Pythagoras, Ptolemy and Archimedes now are.

This knowledge of the field, essential to any worker who hopes to advance human welfare, may be acquired through instruction furnished by schools of all grades, from the highest to the lowest, or it may be a result of observation of the facts, i. e., a fruit of the direct study of the surroundings in which we live.

No one is born with a knowledge of the outer world; this must be acquired solely from experience and toil. Certain natural endowments, memory in particular, favor the acquisition of this necessary knowledge. Many great men have had remarkable memories, Berthelot and President Poincare are outstanding instances. The former knew the titles of more than a thousand of his papers and the volume numbers of the Annales de Chimie in which they were published. It is said that Poincare needed only to write a speech once to know it by heart. He amazed his audience at the Sorbonne by his delivery of a eulogy of the scientific achievements of Berthelot. He spoke more than an hour and a quarter, using no notes or memoranda, and yet with such precision that any one might have thought him an exceptionally well-informed chemist.

The sense of observation is no less valuable; it is essential to the completion of the fragmentary knowledge acquired by the memory during the years of schooling. As we go through life, we are confronted by a multiplicity of facts which demand our attention and efforts. They are so numerous that we cannot expect to learn them from books, and, furthermore, many of them are not common knowledge and consequently cannot be found in courses of instruction. Our knowledge of the world is constantly augmented by the labor of every one of us, but the contributions are extremely unequal varying with aptitude and training. In military instruction, observation is taught by assignments in scouting. The pupil is sent to a given point and on his return he is questioned as to what he saw. Usually, he has seen nothing. He is sent back and told what to look for; the undulations of the ground, the kind of vegetation, isolated trees, hedgerows, roads, houses, the contour of the horizon, and little by little the pupil improves.

High-school exercises in science should be largely planned to develop observational powers. Actual handling of apparatus and materials lends itself excellently to this end. A student is told to heat a material, say iodine, in a test tube, or to dissolve a substance, mercuric sulphate, for instance, in water, and then asked to describe all that he has seen. After he has completed his report, the instructor should point out all that actually can be observed in the experiment.

CONCLUSION

To sum up, we find that the formation of an intellectual elite entails the union of four qualities—industry, imagination, judgment, training. Unfortunately, these qualities are, in a certain degree, contradictory among themselves. The toiler, bound to his task like an ox to a cart, often forgets to pause and meditate; his intellectual activity slows down. The dreamer, the inventor lets himself be guided by his fancies and often lacks common sense. Finally, the abuse of book learning and memory tends to paralyze all the intellectual faculties.

It is extremely difficult to produce a perfect balancing of these diverse faculties. This fact alone is sufficient to explain the infrequency of great men and, *a fortiori*, of men of genius. It is useless to suppose, as many do, that men of genius owe their accomplishments solely to exceptional natural endowments which raise them above the common level. They have qualities which taken singly are not extraordinary, but it is the occurrence of all these qualities in a single mind that is rare.

Most men do not like to work. The suburbs of Paris are filled with small houses inhabited by renters, who perhaps in the beginning worked hard, but only in the hope of soon being able to stop working. Unfortunately, this feeling is all too general in the Latin countries. In 1918, during the Armistice, a party of American engineers who had served as officers were taken for a boat ride down the Seine from Paris to Rouen to see the bridges, locks, etc. To them, the most striking sight was the number of fishermen lining the banks of the river. They could not believe that there were so many Frenchmen content to spend their days watching a cork float on the water, thinking nothing, doing nothing.

Intellectual industry is not much more common than bodily industry. One needs only to scan the interminable lists of candidates for petty government positions which require little if any, mental exertion. The holders of such offices are content to repeat each day exactly what was done the day before. This apathy is apparent from childhood on. How many college students have a horror of real effort? They are satisfied to learn assigned lessons by rote, and never do more than the required amount. Indeed, they often do not try to understand what they learn.

Common sense is perhaps rarer still. In proof of this, consider the results of popular elections. The voters work painfully to make their livings and then cast their ballots for the worst wasters of the public wealth. Industrial conflicts lead to mutual ruination; efforts to squeeze the consumer, who is the goose that produces the golden eggs, bear witness to this same stupidity. Each side devotes itself to efforts to bring about a redistribution of wealth, each hoping to get the bigger share. Neither party comprehends the real problem, which should be its sole interest, i. e., they should both strive for increased production, which would benefit the world at large.

Finally, the question of learning what has been done and the matter of instruction and training is still in a precarious condition. The number of illiterates is still very large. Schools are often more concerned with politics and large enrollments than with the pupils. Unfortunately, this is one of the habitual wastes of democracies. In the higher schools, students obtain cheapened degrees, and higher culture is regarded with more and more contempt. The fetish of equality means a leveling on a lower plane.

Admitting what has been said, let us make a calculation. Suppose one man in ten has a love of industry; one in ten a certain intellectual activity; one in ten common sense; one in ten has been well taught. The probability that these four qualities will be found in one individual will be equal to the fraction 1 over 10 raised to the 4th power, or 1 over 10,000, meaning one in ten thousand—i. e., one in ten thousand may be expected to belong to the intellectual elite. This is not many. The production of a great man requires the union of these same qualities, but each in full bloom. If each quality, developed to this high degree, occurs in one man in a hundred, the probability of their being thus present in a single individual is one in a hundred million. Consequently, we can explain why men of genius are so rare, without seeking the reason in the realms of wonder.

It is a very noble task to aid in the creation of an intellectual aristocracy. The prosperity and lasting glory of any country, all its future welfare, depend on the success of these efforts. Above all, this is a work of education, and no effort should be spared to realize this common good. The family should set a good example to the younger children and inculcate a taste for toil; the secondary school must develop imagination and common sense; and finally the colleges and universities must impart training.

However, an intellectual elite is not the only thing required to make a great nation. There must also be a moral elite, a great class which knows respect for the rights of others, a class to whom the Golden Rule is law, and last but not least, this class also respects and demands respect for real liberty. I do not wish to discuss this last point in detail, for it lies outside my province, and especially because we have a right to expect that the educated classes will have both a highly developed sense of duty and a profound appreciation of independence. It is their duty and privilege to set a noble example to others.

Transcriber’s Note:

Inconsistencies in hyphenation have been standardized.

Minor punctuation errors have been changed without notice.

Spelling was retained as in the original except for the following changes:

Page 15: “by which this prog ress”	to	“by which this progress”
Page 23: “new method of manufactutre”	to	“new method of manufacture”
Page 24: “his plan to an offical”	to	“his plan to an official”
Page 25: “The same Holds true”	to	“The same holds true”
Page 25: “of cast iron and Siemen’s”	to	“of cast iron and Siemens’”

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

> This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, “Information about donations to the Project Gutenberg Literary Archive Foundation.”

- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.

- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.

- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment including outdated equipment. Many small donations ($1 to $5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a